TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Thomas Hutsch	Email	Thomas.hutsch@ifam-dd.fraunhofer.de
Institute Name	Fraunhofer Institute for Manufacturing Technology and Advanced Materials IFAM, Dresden Branch			
Address	Winterbergstr. 28, 01277 Dresden			

2. Technology Overview

Title of Proposed Technology	Development of Advanced Neutron Shielding Composite Material Utilizing Coated Microwires			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), **Next-gen Nuclear Power (X)**, Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Energy & Resources		Mechanical・Materials	
	Medium Category	Small Category	Medium Category	Small Category
	Nuclear Power	Radiation Management Technologies	Metal Materials	Composite Materials
Concept of Proposed Technology	*Describe the core concept, main functions, and potential uses of the proposed technology* The project proposes integrating amorphous metal wire mesh, renowned for its excellent corrosion resistance and mechanical strength, into metal matrix neutron shielding composite materials. This integration aims to overcome the limitations of current shielding materials by combining neutron shielding capacity of heavy metallic base materials with mechanical strengthening effects provided by the fiber reinforcement.			
Rationale for Joint R&D Project	*Describe the need for this joint R&D project and why collaboration is required* Current neutron shielding composite materials used in storage containers for spent nuclear fuel represent strategic resources predominantly controlled by the USA and Japan. Developing independent, reliable shielding solutions is essential for securely managing high-level nuclear waste. Existing materials exhibit limited mechanical properties and reduced longevity. The limitations can be overcome by using composite materials, especially metal matrix fiber composites.			

	The development of such composite materials naturally requires joint efforts of (i) metal wire/fiber specialists providing the reinforcement that is able to significantly improve the mechanical properties while keeping corrosion resistance, radiation tolerance and providing interface strength e.g. by coating and (ii) powder metallurgy specialists creating the matrix and ensuring required interface properties by developing the consolidation and heat treatment processes.
Key Words	*Provide key terms that describe the proposed technology and its main features (min 3, max 5)* 1. Nuclear Shielding 2. Metal matrix composite 3. Metal fiber 4. Wire coating 5. Powder metallurgy

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level 4	End of the Project	Level 6
R&D Objectives	*Describe the specific technologies you aim to develop through this R&D project* The main objective is to establish a scalable production line for fiber reinforced metal matrix composites that can be used as nuclear shielding material, implement robust quality control methods to ensure uniformity and durability, and validate high performance through extensive mechanical and chemical testing Detailed objectives are: - Develop advanced wire twisting and braiding techniques. - Manufacture woven wire materials suitable for composite integration. - Innovate neutron shielding composite materials with superior performance.			
R&D Target Specifications	*Describe the target performance specifications to be achieved through this R&D project, including quantitative values such as measurable parameters* - Microstructure: > 90 % density, homogenous distribution of metal fibers in the matrix with volume content of at least 1 % - Mechanical properties: increased mechanical strength due to fiber reinforcement of > 30 MPa - Neutron shielding effect > 80 % (5mm)			
R&D Plan	*Provide detailed plan outlining the research approach, key tasks, methodologies, expected milestones, and the timeline required to achieve the stated R&D objectives* Stage 1: Microwire Production (6 month) Korea: Produce glass-coated microwires. Germany: Develop specialized wire twisting and weaving processes. Stage 2: Development of Neutron Shielding Composite Materials (20 month) Korea: Provide integrated multi-microwire composite material technologies. Germany: Develop processes for industrial-scale composite material production. Stage 3: Composite Material Verification (10 month) Korea: Manufacture prototypes and perform initial neutron shielding tests. Germany: Conduct durability assessments, comprehensive validation, and performance evaluations.			

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	*Describe the expected roles and responsibilities of the research institute, such as technical contributions, research scope, resources to be provided, and specific tasks to be provided, etc.* - Development of Neutron Shielding Composite Materials using powder metallurgical manufacturing methods - Comprehensive material analysis and evaluation capabilities like SEM analysis, neutron shielding performance assessments, mechanical and thermal property evaluations - Develop processes for industrial-scale composite material production.
Expected R&R of Korean SME Partner	*Describe the expected roles and responsibilities of Korean SME, such as technical contributions, development tasks, testing/validation activities, resource commitments, and collaborations efforts, etc.* - Advanced microwire production technologies, specifically high-boron composition and corrosion-resistant alloys. - Expertise in developing microwire-based weaving methods, including twisting and braiding. - Material analysis and evaluation capabilities including wire mechanical properties testing.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	*Explain the expected outcomes of this joint R&D project, including technological achievements, commercialization potential, and broader industrial or economic impact, etc.* - Establishment and refinement of advanced microwire twisting and braiding techniques. - Composite material with improved corrosion resistance and enhanced durability compared to existing neutron shielding materials. - Expansion of application areas beyond spent nuclear fuel storage casks. - Increased market penetration opportunities through innovative wire processing technologies.

6. References

(1)

(2)

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization